

EduSync Hackathon - Technical Report

Student Submission | TC_Hackathon.pdf | Date: 2026-07-25

PROJECT OVERVIEW:

This document contains the complete technical submission for the TC Hackathon challenge. Our solution focuses on building scalable classroom management tools and real-time student analytics.

KEY FEATURES IMPLEMENTED:

1. Real-time Student & Faculty Synchronization
2. Interactive Coding Sandbox with AI Grading
3. Automated Submission Tracking & Feedback Loop
4. Document & Resource Portal for Classrooms

SYSTEM ARCHITECTURE:

- Frontend: Next.js App Router + React + Tailwind CSS
- Backend: Next.js API Routes + Prisma ORM
- Database: SQLite / PostgreSQL Schema with full multi-role support

CONCLUSION & FUTURE SCOPE:

The project successfully satisfies all hackathon submission criteria and passed unit test suites. Future work includes adding live voice rooms and collaborative whiteboarding.

Submitted by Team Alpha - EduSync Platform